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1: Solving quadratic equations

2: Completing the square

3: Quadratics as functions

4: Quadratic Graphs

5: The Discriminant

6: Modelling with Quadratics

Good question!
There are a number of **practical scenarios** where a quadratic relationship arises between variables. . .

Summations

The sum of 1 to n is $\frac{1}{2}n^2 + \frac{1}{2}n$. I recently had to make a Sports Day spreadsheet which took the number of competing school 'houses' and worked out the total points allocated for an event by finding the formula of the resulting quadratic sequence.

1 competitor: 8 points
2 competitors: 8+7=15
3 competitors: 8+7+6=21
n competitors: $\frac{17}{2}$n−$\frac{1}{2}$n²



Projectile Motion



A projected object, acting only under gravity, follows a *parabolic* trajectory, i.e. its path can be described using a quadratic equation.
In Mechanics you will see the formula: $s = ut + \frac{1}{2}at^2$
Where s is distance/displacement, u is initial speed/velocity, t is time and a is acceleration (in this case gravitational acceleration).
This equation is quadratic in t.

Hannah's sweets

19 There are n sweets in a bag. 6 of the sweets are orange. The rest of the sweets are yellow. Hannah takes at random a sweet from the bag. She eats the sweet. Hannah then takes at random another sweet from the bag. She eats the sweet. The probability that Hannah eats two orange sweets is $\frac{1}{3}$.
(a) Show that $n^2 - n - 90 = 0$.

Quadratic expressions also regularly emerge when there's a product of two expressions involving the same variable (in this case because Hannah ate two orange sweets).

$$x^2 + 5x = 6$$

There are three ways of solving a quadratic equation.
One is by *completing the square*, which we'll do later.
What are the other two ways?

By factorisation

Using the quadratic formula.

Solving without factorising

If the subject only appears once however, it might be easier not to expand out/factorise:

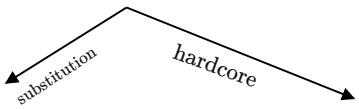
$(x-1)^2 = 5$

If you can't see why the $\pm$ is required, think about the solutions to: $x^2 = 4$.
$2^2 = 4$, but $(-2)^2 = 4$ as well!
So $x = \pm 2$.

Quadratics 'in disguise'

When we have an expression like say $x^2 + 3x - 2$, we say it is "quadratic in x". In trigonometry you will have to solve equations like $(\sin x)^2 + 3\sin x + 2$. We say that the expression is "quadratic in $\sin x$".
Either use a suitable substitution so that you have a 'normal' quadratic, or go straight for the factorisation if you're feeling more confident (recommended!).

Solve $x - 6\sqrt{x} + 8 = 0$



1 Solve $(x+3)^2 = x+5$ using factorisation.

2 Solve $(2x+1)^2 = 5$

3 Solve $\sqrt{x+3} = x-3$

4 Solve $2x + \sqrt{x} - 1 = 0$

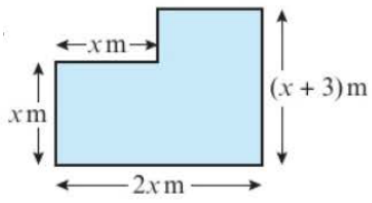
"Use of Technology":
The 'equation' mode on your calculator will solve quadratics in the form $ax^2 + bx + c = 0$. When you're asked for the 'order', use 2 (we'll see why later).

3 Solve the following equations:

i $6x^2 - 7 = 11x$

h $5x^2 - 10x^2 = -7 + x + x^2$

4 This shape has an area of $44\,\text{m}^2$. Find the value of x.



5 Solve the equation $5x + 3 = \sqrt{3x+7}$.

2 Solve the following equations using the quadratic formula. Give your answers to three significant figures.

h $2x^2 + 11x = 5x^2 - 18$

3 For each of the equations below, choose a suitable method and find all of the solutions. Where necessary, give your answers to three significant figures.

e $(2x + 8)^2 = 100$ **f** $6x^2 + 6 = 12x$

Extension: (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1 (i) Use the substitution $\sqrt{x} = y$ $(y \geq 0)$ to find the real root of the equation $x + 3\sqrt{x} - \frac{1}{2} = 0$.

(ii) Find all real roots of the following equations:

(a) $x + 10\sqrt{x + 2} - 22 = 0$;

(b) $x^2 - 4x + \sqrt{2x^2 - 8x - 3} - 9 = 0$.

"Completing the square" means putting a quadratic in the form $(x+a)^2+b$ or $a(x+b)^2+c$

The **underlying reason we do this is because x only appears once in the expression** (e.g. in $(x+2)^2+3$ vs x^2+4x+7), which makes it algebraically easier to handle. This has a number of consequences:

a. Solving Quadratics

If we have a completed square:

$$(x+4)^2-7=0$$

we saw at the start of the chapter how we could rearrange to make x the subject.
Indeed using the quadratic formula is actually solving the quadratic by completing the square – it's just someone has done the work for us already!

b. Sketching Quadratics

We'll see later that if $y=(x+a)^2+b$, then the minimum point is $(-a,b)$

c. In integration

In Further Maths, completing the square allows us to 'integrate' expressions like:

$$\int \frac{1}{x^2-4x+5}\,dx$$

(you will cover integration later this module)

Expand:

$(x+9)^2 = x^2+18$

$(x-5)^2 = x^2-10$

What do you notice about the bold numbers?

The a in $(x+a)^2$ is half the coefficient of x in

Textbook Note:
The textbook uses the formula

$$x^2+bx=\left(x+\tfrac{b}{2}\right)^2-\left(\tfrac{b}{2}\right)^2$$

and similarly

$$ax^2+bx+c=a\left(x+\tfrac{b}{2a}\right)^2+\left(c-\tfrac{b^2}{4a}\right)$$

My personal judgement is that it's not worth memorising these and you should instead think through the steps. Even the textbook agrees!

Therefore if we had x^2+12x, how could we write it in the form $(x+a)^2+b$?

$x^2+12x=(x+6)^2-36$

Since $(x+6)^2=x^2+12x+36$, we want to discard the 36, so 'throw it away' by subtracting.

Further Examples:

$x^2+8x=(x+$

$x^2-2x=(x-$

$x^2-6x+7=$

$=($

Notice that despite the a being negative, we still subtract after the bracket as $(-1)^2$ is positive.

Express $2x^2 + 12x + 7$ in the form $a(x+b)^2 + c$

$= 2\left(x^2 \right.$

$= 2\left((x \right.$

$= 2(x +$

$= 2(x +$

Factorise out coefficient of x^2. You can leave the constant term outside the bracket.

Complete the square inside the bracket (you should have two sets of brackets)

Expand out outer bracket.

Express $5 - 3x^2 + 6x$ in the form $a - b(x+c)^2$

$= -3x^2 +$

$= -3\left(x^2 -\right.$

$= -3\left((x -\right.$

$= -3(x -$

$= 8 - 3(x$

It may help to write in the form $ax^2 + bx + c$ first.

Express $3x^2 - 18x + 4$ in the form $a(x+b)^2 + c$

Express $20x - 5x^2 + 3$ in the form $a - b(x+c)^2$

Solve the equation:

$$3x^2 - 18x + 4 = 0$$

Reflections: Previously we factorised out the 3. This is because $3x^2 - 18x + 4$ on its own is an **expression,** so dividing by 3 (instead of factorising) would change the expression.
However, in an equation, we can divide both sides by 3 without affecting the solutions.

1 Complete the square for the expressions:
a $x^2 + 4x$
b $x^2 - 6x$
c $x^2 - 16x$
d $x^2 + x$
e $x^2 - 14$

2 Complete the square for the expressions:
a $2x^2 + 16x$
b $3x^2 - 24x$
c $5x^2 + 20x$
d $2x^2 - 5x$
e $8x - 2x^2$

4 Given that $x^2 + 3x + 6 = (x + a)^2 + b$ find the values of the constants a and b

5 Write $2 + 0.8\text{x} - 0.04x^2$ in the form $A - B(x + C)^2$ where $A$and$B$are constants to be determined

4 By completing the square, show that the solutions to the equation $x^2 + 2bx + c = 0$ are given by the formula $x = -b \pm \sqrt{b^2 - c}$. (4 marks)

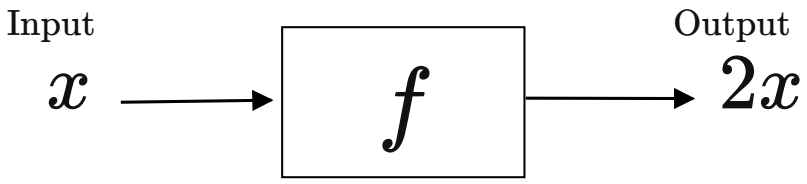
If $ax^2 + bx + c = 0$, prove that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$ax^2 + bx + c = 0$
$x^2 + \frac{b}{a}$
$\left(x + \frac{b}{2a}\right.$
$\left(x + \frac{b}{2a}\right.$
$\left(x + \frac{b}{2a}\right.$
$x + \frac{b}{2a}$
$x = \frac{-b}{}$

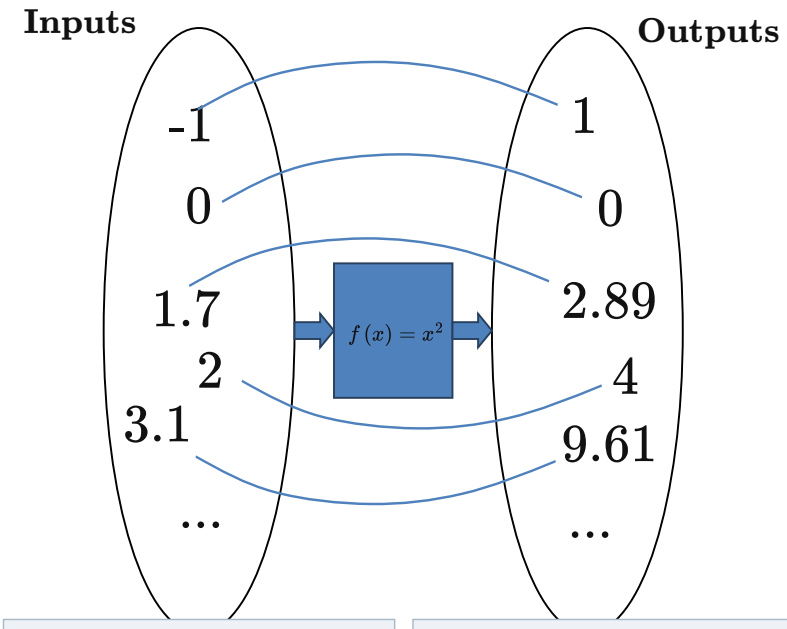
Just use exactly the same method as you usually would!

A function is something which **provides a rule on how to map inputs to outputs.**

We saw at GCSE that functions were a formal way of describing a 'number machine':



You'll cover functions extensively in future chapters, but for now, you need to understand the following concepts:



✎ The **domain** of a function is the set of possible inputs.

✎ The **range** of a function is the set of possible outputs.

The domain of a function could potentially be **any** real number. If so, we'd write:

We might be interested in what inputs x give an output of 0. These are known as the of the function.

✎ The **roots/zeroes** of a function are the values of x for which $f(x) = 0$.

If $f(x) = x^2 - 3x$ and $g(x) = x + 5, \quad x \in \mathbb{R}$

a) Find $f(-4)$

b) Find the values of x for which $f(x) = g(x)$

c) Find the roots of $f(x)$.

d) Find the roots of $g(x)$.

Note: The domain is usually stated for you.

a

b

c

d

Determine the minimum value of the function $f(x) = x^2 - 6x + 2$, and state the value of x for which this minimum occurs.

This means we want to minimise the **output** of the function.

You might try a (clumsy) approach of trying a few values of x and try to see what makes the output as small as possible. . .

But the best way to find the minimum/maximum value of a quadratic is to **complete the square**:

$\mathbf{f(x)}$	Completed square	Min/max value of $\mathbf{f(x)}$	$\mathbf{x}$ for which this min/max occurs
$x^2 + 4x + 9$			
$x^2 - 10x + 21$			
$10 - x^2$			
$8 - x^2 + 6x$			

1 Find the minimum value of $f(x) = 2x^2 + 12x - 5$ and state the value of x for which this occurs.

2 Find the roots of the function $f(x) = 2x^2 + 3x + 1$

3 Find the roots of the function $f(x) = x^4 - x^2 - 6$

2 The function $f(x)$ is defined by $f(x) = x^2 - 2x,\ x \in \mathbb{R}$. Given that $f(a) = 8$, find the two possible values for a.

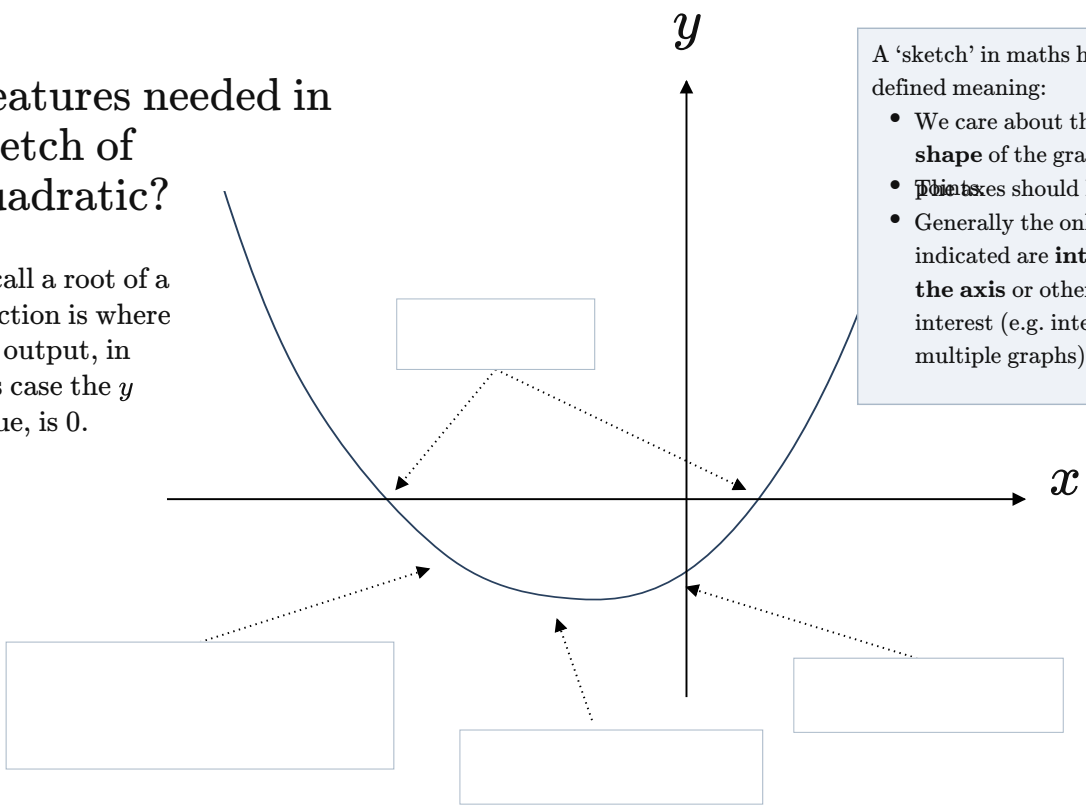
6 The function f is defined as $f(x) = x^2 - 2x + 2,\ x \in \mathbb{R}$.
a Write $f(x)$ in the form $(x + p)^2 + q,$ where p and q are constants to be found.
b Hence, or otherwise, explain why $f(x) > 0$ for all values of $x,$ and find the minimum value of $f(x)$.

8 The function f is defined as $f(x) = 3^{2x} - 28(3^x) + 27,\ x \in \mathbb{R}$.
a Write $f(x)$ in the form $(3^x - a)(3^x - b),$ where a and b are real constants.
b Hence find the two roots of $f(x)$.

Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

Features needed in sketch of quadratic?

Recall a root of a function is where the output, in this case the y value, is 0.



A 'sketch' in maths has a clearly defined meaning:

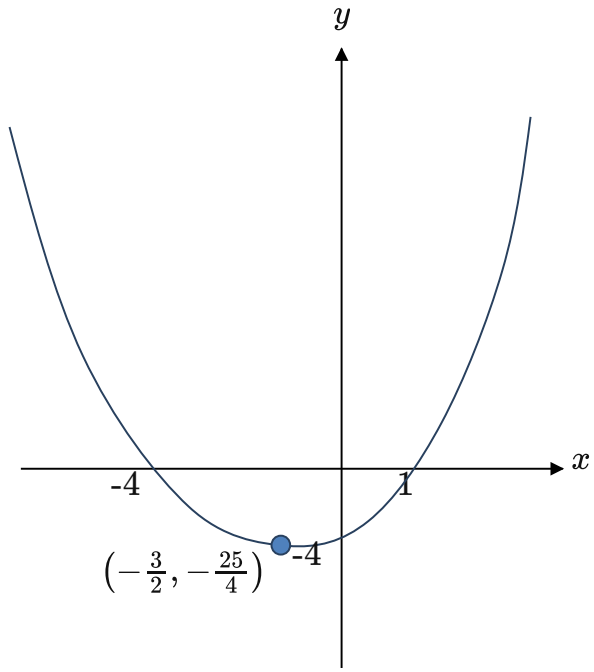
- We care about the **general shape** of the graph, not exact points.
- This axes should have **no scale**.
- Generally the only coordinates indicated are **intercepts with the axis** or other points of interest (e.g. intersections of multiple graphs)

Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

Roots:

y-intercept:

Turning point:



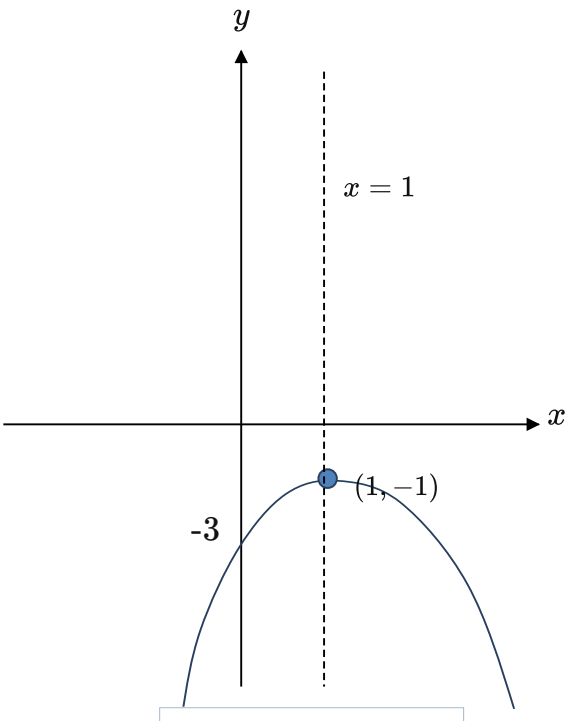
Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

Roots:

y-intercept:

Turning point:



Line of reflection:

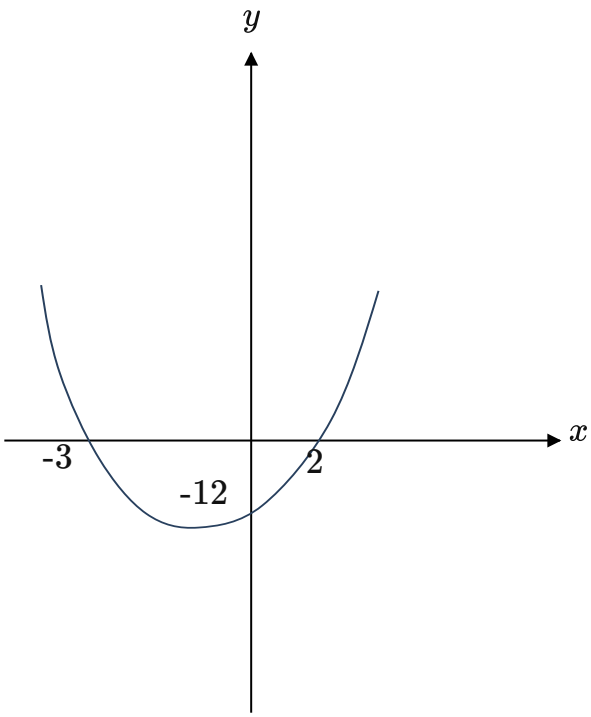
Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

a $y = x^2 + 4$

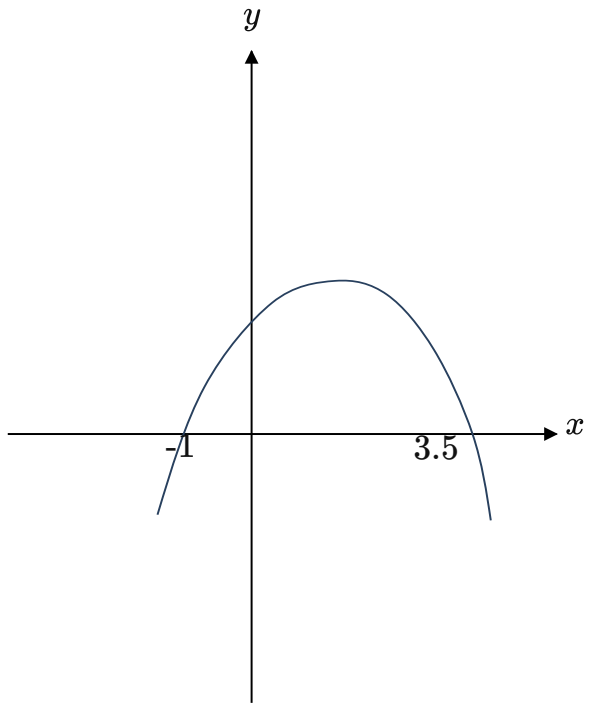
b $y = x^2 - 7x + 10$

c $y = 5x + 3 - 2x^2$

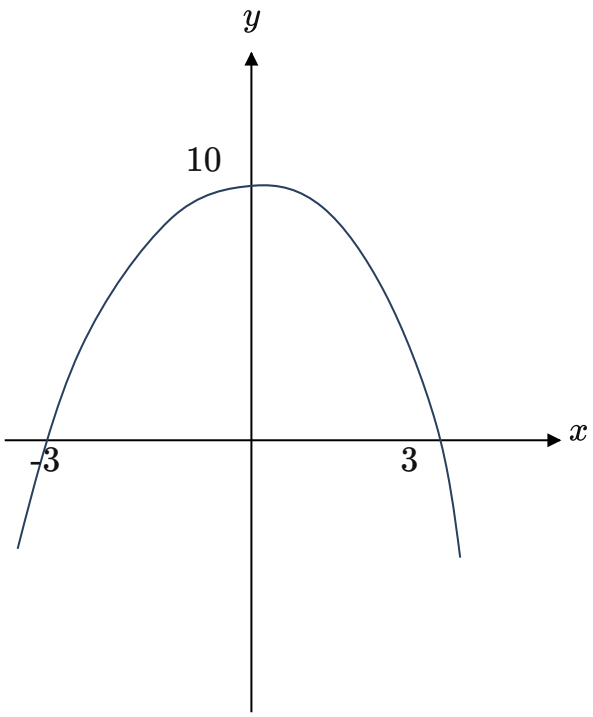
d $y = x^2 + 4x + 11$



Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.



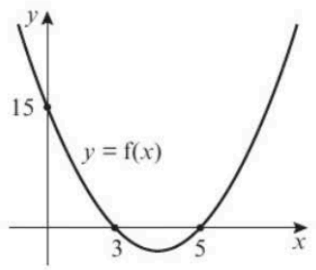
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.



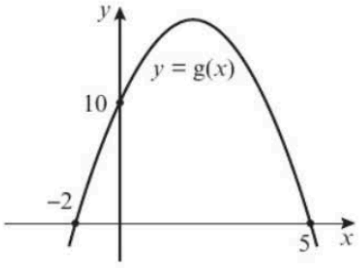
Determine an equation of this quadratic graph.

2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a,b,c for each function.

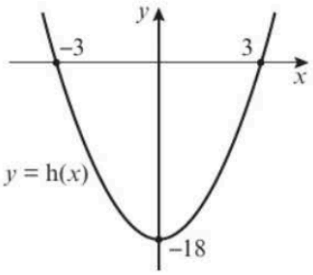
a



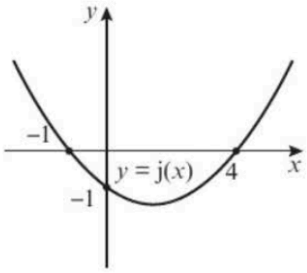
b



c



d



3 The graph of $y = ax^2 + bx + c$ has a minimum at (5,-3) and passes through (4,0). Find the values of a,b,c.

Extension Question:

[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?

$y = x^2$

$y = x^2 - 2x$

$y = x^2 + 2x + 2$

How many **distinct** real solutions do each of the following have?

$x^2 - 12x + 36 = 0$

x=6

$x^2 + x + 3 = 0$

x

W

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$x^2 - 2x - 1 = 0$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

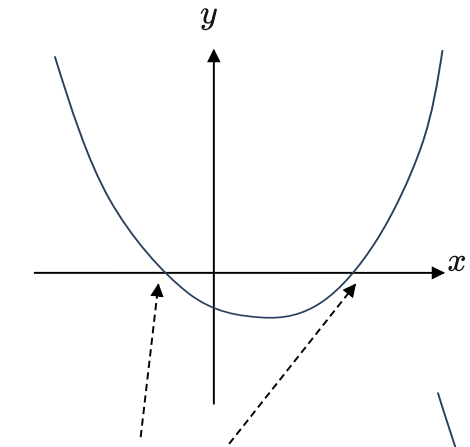
Note: Roots of a **function** f are the values of x such that $f(x) = 0$. Similarly the roots of an **equation** are solutions to an equation in the form $f(x) = 0$

Looking at this formula, when in general do you think we have:

- No real roots?
- Equal roots?
- Two distinct roots?

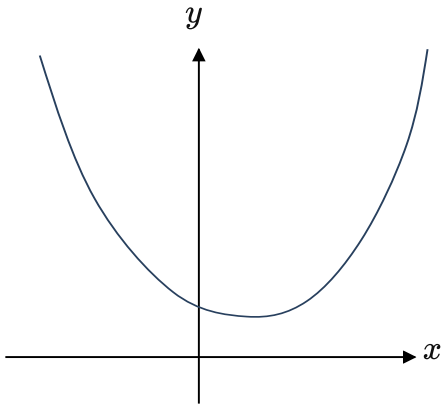
Because adding 0 or subtracting 0 in the quadratic formula gives the same value.

$b2 - 4ac$is known as the discriminant.



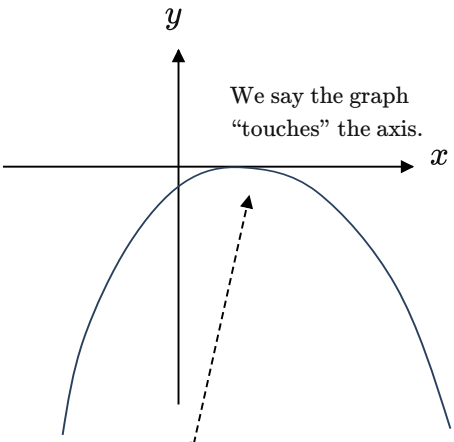
Distinct real roots

$\mathbf{b^2-4ac>0}$



No real roots

$\mathbf{b^2-4ac<0}$



Equal roots

$\mathbf{b^2-4ac=0}$

Why do we say "Equal Roots" not "One root"?

$$x^2 - 12x + 36 = 0$$

Using the quadratic formula gives us the same value in both + and − cases: $x = 6$.

You might wonder why we say "*it has one repeated root*" or "*it has equal roots*", i.e. indicating we have **2 roots** (but with the same value). Why not say it has 1 root?

It is due to the **Fundamental Theorem of Algebra**:

"Every polynomial of order n has exactly n roots."

A **polynomial** is an expression with non-negative integer powers of x, i.e.
$a + bx + cx^2 + dx^3 + \ldots$
All linear, quadratic and cubic expressions are examples of polynomials.

The **order** of a polynomial is its highest power of x. So the order of a quadratic is 2, and a cubic 3.

These roots might be repeated or might not be 'real' roots. $\sqrt{-1}$ is known as a **complex number**, which you will encounter if you do FM. But **it is still a value!**

This theorem means that a quadratic (order 2) will **always** have 2 roots. This is why you should say "no **real** roots" when $b^2 - 4ac < 0$ rather than "no roots", because there are still roots – it's just they're not 'real'! Similarly we must say "*equal roots*" because there are still 2 roots.

Despite the theorem being a simple statement, it was only until 1806 that it was first proven by **Argand**. Clearly by using the quadratic formula we can show a quadratic equation has 2 roots. We can use similar formulas to show that a cubic has 3 roots and a quartic 4 roots. But there is **provably** no such formulae for order 5 (quintics) and beyond. So we have to prove for example that 5 roots exist for a quintic, despite us having no way to find these exact roots!

One side result of the Fundamental Theorem of Algebra is that every polynomial can be written as a product of linear and/or quadratic expressions.

Leibniz claimed in 1702 that a polynomial of the form $x^4 + a^4$ cannot be written in this way. He then got completely burned by Euler in 1742 who managed to do so:

$$x^4 + a^4 = \left(x^2 + a\sqrt{2}x + a^2\right)\left(x^2 - a\sqrt{2}x + a^2\right)$$

There are various other 'Fundamental Laws'. The '**FL of Arithmetic**' you encountered at KS3, which states that "*every positive integer* > 1 *can be written as a product of primes in one way only*". You will encounter the '**FL of Calculus**' in Chapter 13.

Equation	Discriminant	Number of Distinct Real Roots
$x^2 + 3x + 4 = 0$		
$x^2 - 4x + 1 = 0$		
$x^2 - 4x + 4 = 0$		
$2x^2 - 6x - 3 = 0$		
$x - 4 - 3x^2 = 0$		
$1 - x^2 = 0$		

8. The equation $x^2 + 2px + (3p + 4) = 0$, where p is a positive constant, has equal roots.

(a) Find the value of p. (4)

(b) For this value of p, solve the equation $x^2 + 2px + (3p + 4) = 0$. (2)

a) $a = 1$ $b = 2p$, $c = 3p + 4$

$(2p)^2 - 4(1)$

$4p^2 - 12p -$

$p^2 - 3p - 4 =$

$(p + 1)(p -$

$p = 4$

Tip: Always start by writing out a, b and c explicitly.

2 Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions.

3 Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution.

4 Given that the function $f(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s.

5 Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions.

6 The function $g(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.
a Find the value of p.
b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$.

7 $h(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.
a Find the discriminant of $h(x)$ in terms of k.
b Hence or otherwise, prove that $h(x)$ has two distinct real roots for all values of k.

$$x^2 + 5kx + (10k + 5) = 0$$

where k is a constant.
Given that this equation has equal roots, determine the value of k.

Find the range of values of k for which $x^2 + 6x + k = 0$ has two distinct real solutions.

Extension Questions:

1 [MAT 2009 1C] Given a real constant c, the equation

$$x^4 = (x-c)^2$$

Has four real solutions (including possible repeated roots) for:

A) $c \leq \frac{1}{4}$
B) $-\frac{1}{4} \leq c \leq \frac{1}{4}$
C) $c \leq -\frac{1}{4}$
D) all values of c

Square rooting: $x^2 = \pm(x-c)$

2

3 [MAT 2011 1B] A rectangle has perimeter P and area A. The values P and A must satisfy:

A) $P^3 > A$
B) $A^2 > 2P + 1$
C) $P^2 \geq 16A$
D) $PA > A + P$

Let x and y be the width and height. Then A=xy,P=2x+2y. Substituting:

The new A Level has a particular emphasis of the application of theory to real-life situations. A mathematical model is the maths used to model such a situation, possibly with some simplifying assumptions.

Example (from textbook): A spear is thrown over level ground from the top of a tower. The height, in metres, of the spear above the ground after t seconds is modelled by the function: $h(t) = 12.25 + 14.7t - 4.9t^2, \quad t \geq 0$

a) Interpret the meaning of the constant term 12.25 in the model.
b) After how many seconds does the spear hit the ground?
c) Write $h(t)$ in the form $A - B(t-C)^2$, where A, B and C are constants to be found.
d) Using your answer to part c or otherwise, find the maximum height of the spear above the ground, and the time at which this maximum height is reached?

a

$h(t) = 0 \qquad 12.25 + 14.7t - 4.9t^2$

b

$23.275 - 4.9(t-1.5)^2$

c

d

2 A car manufacturer uses a model to predict the fuel consumption, y miles per gallon (mpg), for a specific model of car travelling at a speed of x mph. $y = -0.01x^2 + 0.975x + 16,\ x > 0.$

a Use the model to find two speeds at which the car has a fuel consumption of 32.5 mpg.

b Rewrite y in the form $A - B(x - C)^2,$ where A, B, C are constants to be found.

c Using your answer to part b, find the speed at which the car has the greatest fuel efficiency.

d Use the model to calculate the fuel consumption of a car travelling at 120 mph.
Comment on the validity of using this model for very high speeds.

A fertiliser company uses a model to determine how the amount of fertiliser used, f kilograms per hectare, affects the grain yield g, measured in tonnes per hectare. $g = 6 + 0.03f - 0.00006f^2.$

According to the model, how much grain would each hectare yield without any fertiliser? (1 mark)

One farmer currently uses 20 kilograms of fertiliser per hectare. How much more fertiliser would he need to use to increase his grain yield by 1 tonne per hectare? (4 marks)

4 A football stadium has 25000 seats. The football club know from past experience that they will sell only 10000 tickets if each ticket costs \pounds30. They also expect to sell 1000 more tickets every time the price goes down by \pounds1.

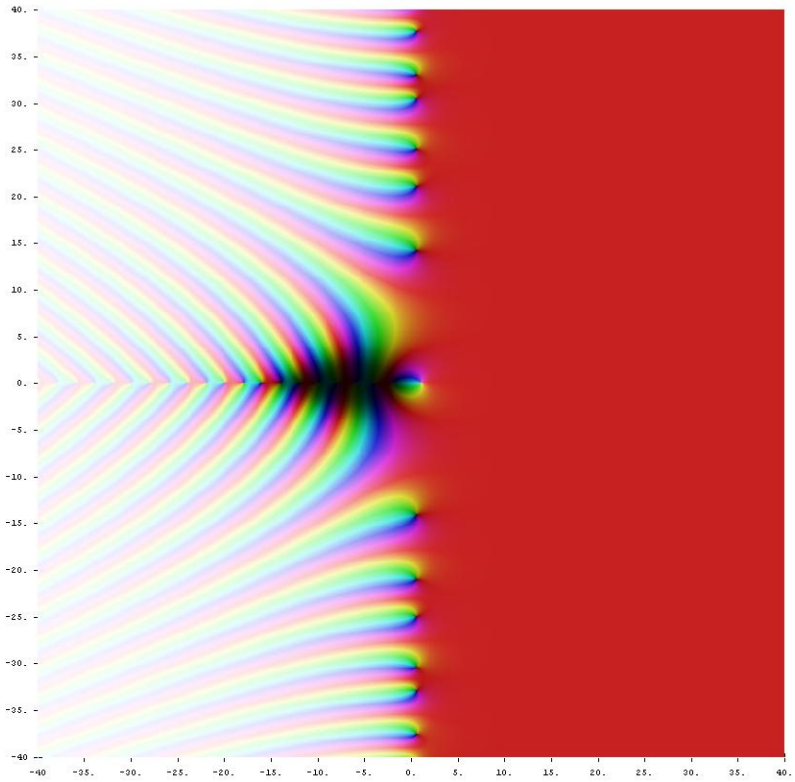
a) The number of tickets sold t can be modelled by $t = M - 1000p$, where \pounds$p$ is the price of each ticket and M is a constant. Find M.

The total revenue, \pounds$r$, is the number of tickets sold multiplied by the price of each ticket: $r = p(M - 1000p)$.

b) Rearrange r into the form $A - B(p - C)^2$, where A, B, C are constants to be found.

c) Using your answer to part b or otherwise, work out how much the football club should charge for each ticket to make the maximum amount of money.

We saw earlier that the roots of a function f are the values x such that $f(x) = 0$.



The **Riemann Zeta Function** is a function that allows you to do the infinite sum of powers of reciprocals, e.g.

$$\zeta(2) = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots$$
$$\zeta(3) = \frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \ldots$$

One of the 8 '**Clay Millennium Problems**' (for which solving any attracts a $1,000,000 prize) is to **showing all roots of this function have some particular form**, i.e. the form of x such that $\zeta(x) = 0$.